

nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D•09

Concept Diagrams, Sections & Elevations

№10 Diagrams

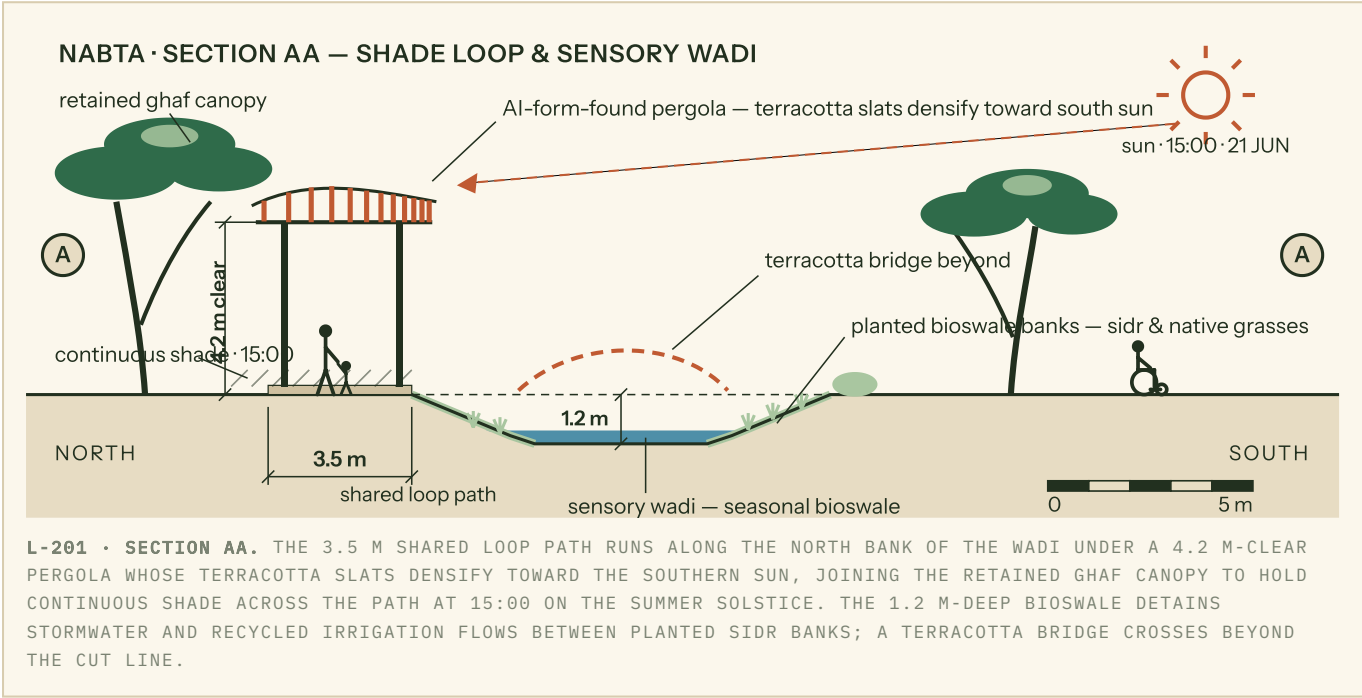
№11 Sections



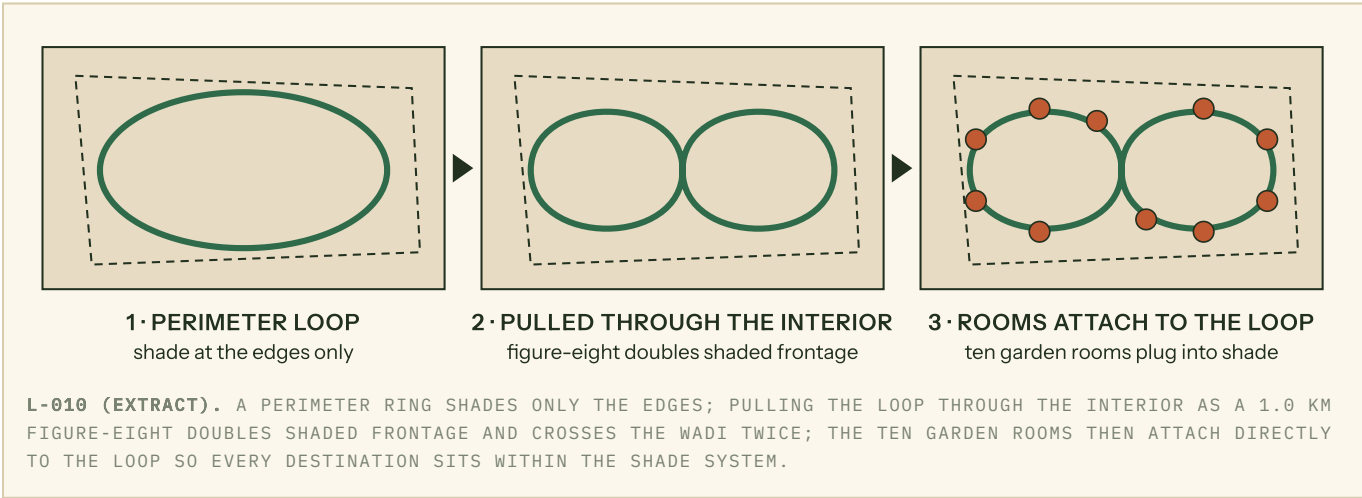
Package 9 – Concept Diagrams, Preliminary Design Drawings, Sections & Elevations

Deliverable 10 **Deliverable 11** This package presents the graphic core of the «نبطة» **NABTA – Wellness Park** entry for Al Sufa 2 (15,000 m²), under the concept **The Living Shade Loop**: the primary illustrative section through the shade loop and sensory wadi, the three-step concept evolution diagram, and the register of preliminary design drawings issued with the full submission. All drawings derive from a single federated CAD/BIM model, as required by the brief; the section below is reproduced at reading scale from drawing L-201.

SECTION AA – THROUGH THE SHADE LOOP AND SENSORY WADI (1:200 @ A2, REPRODUCED AT READING SCALE)



CONCEPT EVOLUTION – FROM RING TO LIVING LOOP



DRAWING REGISTER – PRELIMINARY DESIGN SET

The full entry includes the following drawings, all cut from one federated Revit/Rhino model (shared site coordinates, IFC 4 exchange, ISO 19650 naming) and issued as DWG + PDF per the brief's CAD/BIM

requirement. Sections and elevations reference the common ± 0.000 park datum shown dashed in Section AA.

NO.	TITLE	SCALE	FORMAT
L-001	Illustrative masterplan	1:500	A0 · DWG/PDF from BIM
L-010	Concept & systems diagrams (shade, water, sensing)	NTS	A3 · PDF
L-101	Shade loop & garden rooms — general arrangement	1:250	A1 · DWG
L-201	Section AA — shade loop & sensory wadi	1:200	A2 · DWG/PDF
L-202	Section BB — play dune, majlis plaza & café terrace	1:200	A2 · DWG
L-301	Pergola elevations, north & south faces	1:100	A1 · DWG from Revit (LOD 300)
L-302	Majlis plaza & café terrace elevations	1:100	A1 · DWG
L-401	Pergola node & slat-fixing details	1:20	A3 · DWG
L-402	Bioswale edge, weir & terracotta bridge details	1:20	A3 · DWG
L-501	Planting & irrigation zoning plan	1:250	A1 · DWG + GIS
L-601	AI form-finding record — inputs, objectives, selected generation	NTS	A3 · PDF + parametric definition

HOW THE PERGOLA WAS FORM-FOUND

The pergola geometry in Section AA is not drawn — it is computed, then judged by the design team. A notional canopy surface over the loop centreline was meshed and exposed to an annual solar simulation at the Al Safa 2 coordinates. The mapping is direct: **simulated solar load per mesh face → slat density**. The most irradiated faces receive slats at 180 mm centres, tapering linearly to 600 mm centres on the sheltered north edge — the south-densifying gradient visible in the section.

An evolutionary solver then optimised slat spacing, chord curvature and post positions against a single primary target — **continuous shade along the 3.5 m path centreline at 15:00 on the summer solstice**, the governing heat-stress hour drawn in Section AA — with minimum slat material and preserved low-angle December sun as secondary objectives. The team reviewed the leading generations against the 4.2 m clear height and sightlines to the 1.2 m wadi before selecting one; that decision, and the solver record, are documented on sheet L-601. The selected generation exports as native parametric geometry alongside the IFC model, so the same definition regenerates fabrication data at detailed design. The result is a roof that visibly "reads" the sun — denser where Dubai's sky is harshest, open where the retained ghaf canopy already does the work.